Hwang Doyeon [pdf]

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ABOUT

I am a software engineer who likes to understand a domain and then turn it into a product. I have worked in carbon accounting, clinical trials and B2B SaaS, building software that people actually use alongside colleagues who are not developers.

What to build matters, and so does how to run it. I have designed and improved the path from a commit to a user who depends on it, and lately I focus on using AI to close the gap between people and the software they operate.

EXPERIENCE

RenewEarth Lab (CarbonLink)

2024.09 — present

Tech Manager · Full-stack · Infrastructure · Developer productivity

I joined as one of three founding members and the first engineer — no product, no code, no infrastructure. I built the carbon accounting platform CarbonLink from zero: the design, the calculation engine, the web apps, the async workers and the AWS infrastructure. In 2026, as the team grew, I widened my scope to the engineering baseline — CI/CD, security guardrails and the written record of decisions.

2024.09 — **CarbonLink** — corporate carbon inventory platform

- **GHG Protocol Scope 1·2·3 emissions** — implemented the Scope 3 calculation methods for Categories 1·2·3·4·6·7·8·9·13 myself, and cleaned and applied emission-factor databases (IEA, Ecoinvent, Exiobase).
- **One schema for messy uploads** — Excel, document OCR and SAP ERP land in the same model; 66 million SAP rows across 8 object types were migrated with their original IDs and reconciled by count.
- **Heavy work moved to queues and workers** — recalculation, PDF reports and summary aggregation run in separate workers, and services written in different languages talk over Redis Streams.

2025 — **Audit readiness** — audit trail

- **Third-party verification views** — a drive-style workpaper browser, year-over-year comparison and Scope 1·2 activity-data sheets, shipped to the development environment.
- **Measuring the audit trail** — I turned a ten-question auditor set into system conditions and raised the share of data-changing points that leave a record from 7.3% to 93.9% (865 places), and the share that record who made the change from 1.1% to 99.2%.

2025 — **Infrastructure & delivery**

- **13 AWS ECS services managed with Terraform** — Route53–ALB–ECS, per-service auto scaling, and Redis Streams·Celery·BullMQ queues separated by language.
- **Self-hosted GitHub Actions runners** — cut CI queueing from over 40 minutes to 16 seconds and raised concurrent jobs from 2 to 5, with a dedicated account, automatic restarts and resource caps.

2026 — **relay** — team collaboration record

- **Async work as a git-based signal stream** — rules are enforced by automated checks rather than documents, so a PR that only adds a signal merges without human hands once the checks pass.
- **Weekly summaries generated automatically** — teammates who never read the code can still follow progress, decisions and blockers; 33 planning documents moved from Notion to GitHub Projects so the board and the record each have one job.
- **An AI reviewer and a macOS app** — a reviewer that flags records which pass the format rules but cannot be read, plus a SwiftUI macOS app over the same data, built without external libraries and shipped to 1.10.0.

Mediaiplus Inc.

2022.02 — 2024.03

Developer

- **Clinical-trial data integration** — collected and normalised worldwide clinical-trial data into a single big-data structure. Institution-specific formats and schemas were unified with Python·MySQL pipelines and refreshed daily.
- **Legacy migration** — moved a Tomcat + Spring Boot service to React and Express, added React Chart visualisation, improved search speed with TanStack Query, and wired JWT authentication and automated AWS RDS backups.
- **Deployment automation** — managed delivery as code (IaC) with Jenkins·Terraform·Ansible behind an AWS Load Balancer.
- **Regulation & guideline service** — built the web service for clinical-trial regulations and guidelines, and self-hosted the open-source Vicuna model behind an OpenAI-compatible API for prompt-engineered data cleaning.

PUBLICATIONS

How AI Research Views Dance: A Topic Modeling Analysis of Domestic and International Scholarly Index Databases

Kim Y., Hwang D. · Journal of Digital Culture Archive (JDCA), KCI, 2026

Topic modelling on AI-and-dance research indexed in SCOPUS and KCI, comparing the knowledge structures of domestic and international work.

Theme Types and Similar Expression Techniques of Korean Gyobang Dance Based on YouTube Data

Kim Y., Hwang D. · Journal of Digital Culture Archive (JDCA), KCI, 2024

Collected Gyobang dance video metadata through the YouTube Data API, then analysed theme types and audience perception with sentiment analysis and topic modelling.

TECHNICAL SKILLS

TypeScript, Python, JavaScript · React, Next.js, Express, FastAPI · Docker, Terraform, Ansible, AWS · MongoDB, PostgreSQL, MySQL · GitHub Actions, Jenkins

EDUCATION

Hansei Cyber Security High School — Hacking & Security

2019 — 2022

CERTIFICATIONS

Information Processing Technician (정보처리기능사)

Human Resources Development Service of Korea